

Irmak Ergin

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PhD Candidate - Computational Cognitive Neuroscience

Gwilliams Lab of Speech Neuroscience & Neural Prosthetics Translational Lab, Stanford University

Education

Stanford University, USA

Ph.D. in Psychology – Neuroscience Track

2023–Present

Supervisor: Dr. Laura Gwilliams

3rd year, Anticipated graduation date: 06/2028

University of Trento (CIMEC) / SISSA, Italy

M.Sc. in Cognitive Science – Neuroscience Track

2021–2023

Grade: 110/110

Supervisors: Dr. Daniel Baldauf & Dr. Davide Crepaldi

Bogazici University, Turkey

B.A. in Psychology

2015–2021

Honors

Supervisors: Dr. Resit Canbeyli & Dr. Aysecan Boduroglu

Research Experience

PhD Candidate, Stanford University

Palo Alto, CA

2021–2023

- In collaboration with Neural Prosthetics Translational Lab (NPTL)- BrainGate: Investigating neural mechanisms of speech processing by analyzing rare large-scale single-neuron Utah array recordings obtained from human motor cortex and Broca's area from a BCI patient. This work leverages state-of-the-art speech and language models, using embeddings and mapping methods for spatio-temporal neural analysis.
- Designed, developed, and validated a novel scanner-compatible slider device that synchronizes with neural and behavioral recording software, enabling millisecond-resolution simultaneous measurement of physiology and behavior.

MSc Researcher, Center for Mind/Brain Sciences

Trento, Italy

2023–Present

- Analyzing MEG data to study neural synchronization and connectivity in auditory attention and object recognition; conducted data preprocessing, ROI detection, and connectivity analysis for linguistic and environmental auditory stimuli.

Intern, NYU-MPI Center for Language Music and Emotion (CLaME)

NYC, USA

Summer 2021

- Applied timeseries analysis (e.g., Phase Locking Analysis) to EEG data to investigate neural mechanisms underlying musical tension.

Technical Skills

Programming: Python, MATLAB, R

Neuroimaging Methods: Utah Arrays, MEG, EEG

Computational Neuroscience & Signal Processing: dimensionality reduction, cross- validated regression, noise reduction, Fourier transform, source localization

Machine Learning & Data Analysis: scikit-learn, SciPy, NumPy, Pandas, librosa, pysptk, Brainstorm, FieldTrip, Psychtoolbox

Deep Learning Frameworks: PyTorch

Visualization: Matplotlib, Seaborn, Plotly

Tools: Git, Jupyter Notebook

Awards and Scholarships

Stanford Human-Centered AI & Data Science Fellowship	2025–2027
Stanford Center for Mind, Brain, Computation, & Technology Fellowship	2024–2026
Fulbright Graduate Degree Scholarship <i>[declined]</i>	2023–2026
UniTrento-SISSA Master's Degree Scholarship	2021–2023
German Academic Exchange Service (DAAD) Scholarship <i>[declined]</i>	2021–2023
TÜBİTAK National Scholarship for Undergraduate Researchers	2021
European Union Erasmus+ Grant	2020

Publications

Ergin, I., Kazemian, A., Rojas, E., Hahn, N., Kamdar, F., Vo, C. K., Madugula, S. S., Wang, R. Z., Fan, C., Kunz, E., Avasino, D., Hochberg, L., Henderson, J. M., Willett, F., & Gwilliams, L. Lexical representations are sequenced through a micro-scale dynamic code in inferior frontal gyrus. *Cognitive Computational Neuroscience Conference 2026*.

Ergin, I., Kries, J., Gupta, S., Papworth Burrell, M., & Gwilliams, L. (2024). Measuring Naturalistic Speech Comprehension in Real Time. *Behavior Research Methods*, 58(4), 93. <https://doi.org/10.3758/s13428-026-02941-1> [\[pdf\]](#)

Boyce, V., Prystawski, B., Abutto, A. B., Chen, E. M., Chen, Z., Chiu, H., **Ergin, I.**, ..., Frank, M. C. (2024). *Estimating the replicability of psychology experiments after an initial failure to replicate*. [\[pdf\]](#)

Conference Poster Presentations

Ergin, I., Kries, J., Gupta, S., & Gwilliams, L. (2024). Introducing A Real-Time Measure of Comprehension During Natural Speech Listening. *Society of Neurobiology of Language*, Brisbane, Australia.

Ergin, I., Padovani, F., Marelli, M., Crepaldi, D., & Ktori, M. (2024). Morpheme Positional Constraints and Grammatical Class. *California Meeting on Psycholinguistics (CAMP)*, Stanford, USA.

Ergin, I., Canbeyli, R. (2021). Associations in Response to Compound and Reversible Words in Bilinguals. *49th Meeting of the European Brain and Behavior Society*, Lausanne, Switzerland.

Relevant PhD Coursework

CS 224S: Spoken Language Processing

CS 224N: Natural Language Processing with Deep Learning

NEPR 209: Mathematical Tools in Neuroscience

Teaching

TA, Introduction to Cognitive Neuroscience

Winter 2026

with Prof. Justin Gardner

TA & Guest Lecturer, Experimental Methods

Fall 2025

with Prof. Michael Frank

TA & Guest Lecturer, Data Science for Neuroscience

Winter 2025

with Prof. Laura Gwilliams

TA & Guest Lecturer, Introduction to Psycholinguistics

Fall 2024

with Prof. Cory Shain